BOOK PROSPECTUS · OCTOBER 2026

Using Data to Drive *Business Performance*

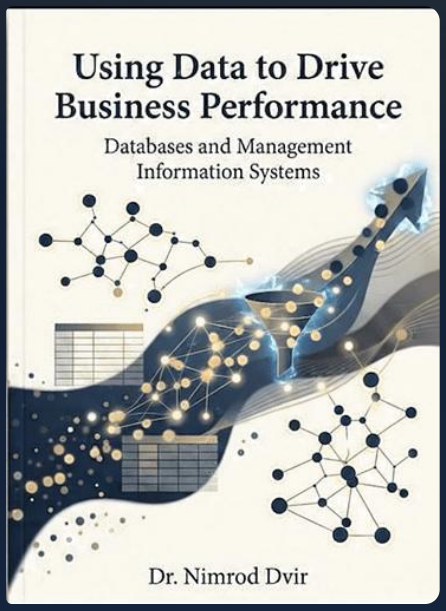
Databases and Management Information Systems

Nimrod Dvir, PhD
Massry School of Business · University at Albany, SUNY

The two-page brief · PDF ↗

Open the live reader of the current version ↗

Prepared for Cengage · Publisher discussion draft · October 2026



Where the courseware stands today

Written, taught and revised in the classroom.

VERSION 1 · SUMMER 2026

Tested in class

Tested with a small group of students, then revised where they needed more help.

VERSION 2 · FALL 2026

In use with 500 students

The full interactive textbook, labs and quizzes, live this semester.

Explore the live courseware ↗

NEXT EDITION · PROPOSED

16 chapters, built on research

The edition this page proposes, to develop together with Cengage.

See the proposed edition ↓

A GUIDE TO THIS PAGE

What's included?

This page describes the book and its development, and the research behind it: five studies, from the market to my own classroom. It then sets out the proposed edition, the full research library, and the questions to decide together.

01 **Overview of the book**
What it teaches, who it is for, how it differs, and the opportunity

02 **How the book was built and tested**
From seven years of teaching to two versions tested in class

03 **The research behind the courseware**
Five studies, each with its own findings and conclusion

- 01 Market and adoption
- 02 Syllabi and institutions
- 03 Employment and industry
- 04 Textbooks
- 05 Classroom

04 **The proposed edition**
A working proposal: 16 chapters in five parts, cases, routes and a teaching schedule

05 **The full research library**
Current studies and earlier work, every PDF in one place

06 **Open questions and next steps**
Three areas to decide together, and how to move forward

Part one

The book

Courseware about the relationship between data and business performance, already in use, and ready for its next edition.

THE PROPOSITION

What is this book about?

This is courseware about the relationship between data and business performance. Students learn how the way they organize, query and present data changes the decisions a business can make.

It brings together a textbook, an interactive reader, guided practice, labs, auto-graded quizzes and a semester project, designed to work as one course for undergraduate business students.

I want students to see the whole journey: understand the data, build a reliable database, ask a useful question in SQL, check the answer, and turn it into a recommendation. AI is part of that journey, with students responsible for checking and explaining the work.

READERS AND ADOPTERS

Who it is for

Students

Business majors who may never have planned to write code. Each chapter explains why a step matters before asking for it, and asks students to check and explain their work.

Instructors

A predictable chapter package: reading, a guided build, an interactive lab, an auto-graded quiz and instructor materials. Less stitching together of sources, more teaching.

Programs

One integrated course instead of a theory text plus a separate software manual, with a full MIS route and a shorter business-core route through the same chapters.

WHY THIS BOOK IS DIFFERENT

A database course that finally feels like *business*

Most database textbooks open with theory and diagrams, and hope students stay long enough to see why it matters. This one opens with the questions a manager actually asks.

01 **Starting with the business**
Chapter 1 opens with performance, goals and KPIs, before any database theory.

02 **Teaching data before databases**
Data types, measurement scales and organizing information come first.

03 **Using a database before designing one**
Students query a supplied database in Chapters 4 to 6, then design their own.

04 **Ending with the decision**
Analytics, dashboards and a full Power BI chapter lead to a decision memo.

05 **One SQL dialect, flexible tools**
SQL in the PostgreSQL dialect, with a portable core and explicit dialect differences. The Access route is in use today; a MySQL route is planned and will be tested on the same cases.

06 **Industry tools students can keep**
Supabase hosts the PostgreSQL project database; Power BI is the main reporting studio, with Tableau as an alternate route.

07 **Making AI a habit**
From Chapter 5 on, an AI verification asks students to accept, revise or reject AI-assisted work and to take responsibility for the result.

08 **Practice in every chapter**
A Let's Build, an interactive lab and an auto-graded quiz, all on the book's own cases.

09 **Proven in class**
Version 1 was tested in summer 2026; version 2 is in use with 500 students this fall. **See the classroom research**

WHAT I AM PROPOSING WE DEVELOP TOGETHER

The opportunity

The starting point is a complete course already in use. The opportunity is to develop it with editorial support, a clear tool pathway, and wider instructor review.

Ready asset	The interactive textbook, chapter practice, quizzes and instructor resources are in use with 500 students this semester. The work ahead is revision and review, not a first draft.
What we would develop	The research-based 16-chapter edition, running cases tested with reviewers and adopting instructors, and tool routes tested on those cases: PostgreSQL with PGlite and Supabase as the foundation, the Access route in use today, and a planned MySQL route.
Two course routes	A full applied MIS database route and a shorter business-core route, selected from the same chapters.
An open slot	No title appeared at more than 3 of the 31 schools we reviewed, and Cengage's applied title in this space, *Concepts of Database Management*, was last revised in 2021. This book can serve the applied course with current SQL, analytics and AI, and complement deeper texts such as *Database Systems.*
Portfolio fit	A business-focused pathway from data meaning to SQL, reporting and a decision memo. Positioning alongside existing database titles is a conversation to have together.

Known partner	I already work with Cengage as a member of its Intro Computing Thought Leadership Board.
More opportunities	Selected modules may also fit accounting information systems and analytics foundations. Existing Access materials and a planned MySQL route can support different teaching environments.

THE AUTHOR'S STARTING POINT

Why and how it was built

After seven years of teaching this course to more than 4,000 students, I had a familiar problem: I liked parts of several textbooks, but no single book brought together everything I wanted students to learn. I often assigned two books and wrote the missing pieces myself.

I also saw a focused opportunity. In one published study, 85% of 61 business schools required at least one information systems course in the business core, and in our review no textbook appeared at more than 3 of the 31 schools. The research below documents both.

So I began building the courseware I wanted to teach. Over a year of writing and work with colleagues, it became a complete course. The classroom gave me the starting point. The research helped me question and refine it.

What shaped the design

Seven years in the classroom

Teaching the course to more than 4,000 students, and seeing where business students need more context, practice or confidence.

Ten years of editing

More than ten years as a magazine editor: clear, visual writing, organized around the reader.

Five studies

Market, syllabus, employer, textbook and classroom research. **See the research**

Quick iteration

Because I built the platform, tools, examples and AI guidance can change each semester instead of each edition.

How the design changed, and where it was tested

1

SEVEN YEARS OF TEACHING

Two books and my own materials

The gaps I filled each semester became the outline of the courseware.

2

A YEAR OF WRITING

A complete course

Textbook, interactive reader, Let's Builds, labs and quizzes, written with colleagues.

3

SUMMER 2026

Version 1, tested in class

A small group of students used the full course. I revised the chapters where they needed more help.

4

FALL 2026

Version 2, in class with 500 students

The revised courseware is in use this semester, with student accounts, automatic quiz grading and an instructor dashboard.

5

NEXT

The research-based edition

Five studies tested and refined the design. The result is the 16-chapter edition proposed below.

Ready now

- The full textbook, live as an interactive digital edition and in use at the University at Albany
- A Let’s Build, an interactive lab and an auto-graded quiz in every chapter
- Student accounts, automatic quiz grading and an instructor dashboard
- Two running cases: the Grading Database and PetVax
- An Access route: Access labs and an Access appendix

Open the live courseware ↗

Planned for the next edition

- The research-based 16-chapter outline
- In-browser PostgreSQL (PGlite) practice workspaces inside the reader
- A planned MySQL route, to be tested on the shared cases
- Automated feedback on SQL and lab answers (prototype in testing)
- A small retail business, proposed as an alternative case for evaluation

Part two

THE RESEARCH BEHIND THE COURSEWARE

Five studies. One book.

Before proposing an edition, I tested the idea from five directions: the market, the courses, the employers, the textbooks and my own classroom. Each study below has its method, its findings and a main conclusion.

Research 01

Market and adoption

375,418

business bachelor's degrees a year in the U.S.

Research 02

Syllabi and institutions

63

course documents from 31 institutions

Research 03

Employment and industry

52.9%

of data analyst job postings ask for SQL

Research 04

Textbooks

60

titles catalogued; four compared chapter by chapter

Research 05

Classroom

About 600

students in summer and fall 2026

01

RESEARCH 01 · MARKET AND ADOPTION

Market and adoption research

How many students reach a course this book can serve, and where could it be adopted?

Full market research · PDF ↗

How the market research was done

What we reviewed	NCES degree statistics, IPEDS completions through Data USA, published curriculum studies, and AACSB accreditation data. Syllabus, textbook and employer studies follow as their own research.
How	We retained published figures with their source years, separated degree counts from course and program evidence, and combined two of them into illustrative reach proxies. Each proxy is shown with its assumptions and would need publisher validation.
What we looked for	The scale of the student audience, where relevant courses are required, and which course settings are plausible adoption opportunities.

FINDING 1

A large, required course

375,418
DEGREES, 2021–22

Business bachelor's degrees awarded in the U.S.

About one in five of all bachelor's degrees.

Source: NCES [1]

85%
OF 61 BUSINESS SCHOOLS

Require at least one information systems course in the business core

52 of 61 schools studied (89% of them AACSB-accredited); 3% required two IS courses.

Source: Fustos, Morris and Haga [2]

12,092
AWARDS IN 2024

Management information systems degrees awarded

Up 3.4% over the prior year. Counts all award levels reported, not only bachelor's degrees.

Source: Data USA [3]

87.3%
OF 394 IS PROGRAMS

Offer a data and information management course

344 of 394 U.S. undergraduate IS programs in business schools (data collected in 2013).

Source: Hwang, Ma and Wang [4]

285 of 535
AACSB SCHOOLS

U.S. AACSB-accredited schools that offer an information systems program

A further 109 offer an undergraduate business analytics program (Table 2).

Source: Zheng et al. [5]

82%
OF 74 IS MASTER'S PROGRAMS

Require a data, information and content management course

61 of 74 programs at AACSB schools: a graduate audience for the same material.

Source: Yang [6]

81,835
AWARDS IN 2024

Accounting degrees awarded, an adjacent audience

Accounting information systems courses cover much of the same ground. All award levels reported.

Source: Data USA [7]

1,098
INSTITUTIONS

AACSB-accredited institutions worldwide

In 71 countries and territories, as of September 2026.

Source: AACSB [8]

FINDING 2

Two routes, one book

A commonality we found: *most business schools in the published studies require an information systems course of every business student, and most IS programs offer a database course.*

The proposed routes address these two settings. Fit must be reviewed against each course's outcomes and available teaching time. In round numbers:

About 320,000
ILLUSTRATIVE REACH PROXY

Business students who reach a required IS course

375,418 business bachelor's degrees (2021–22) × the 85% of business schools that require at least one IS course (2017 study).

Source: Our estimate from [1], [2]

About 10,600
ILLUSTRATIVE REACH PROXY

MIS students in programs with a database course

12,092 MIS degrees across all award levels (2024) × the 87.3% of IS programs that offer the course (2015 study).

Source: Our estimate from [3], [4]

Planning assumptions: each proxy multiplies a degree count by a historical, program-level share from a different source year. They are not enrollment-weighted course-seat estimates, and the MIS count covers all award levels. The two audiences overlap and must not be added together. Publisher enrollment and adoption data are needed to estimate addressable annual seats.

FINDING 3

Adoption fit

Which courses could use the courseware, what they need, and what a review should confirm.

SEGMENT	LIKELY NEED	VALIDATION FOCUS	ADOPTION OPPORTUNITY
Primary Applied business database / MIS data management	Relational thinking, SQL, design, implementation, and business interpretation in one coherent learning path.	Course outcomes, annual sections and student seats, required database engine, incumbent materials, and switching criteria.	Full courseware adoption with readings, guided practice, independent labs, assessment, and instructor resources.
Business core data / IS course	Data meaning, information systems, queries, reporting, and a bounded project.	Which chapters fit the existing course and how much SQL and design it expects.	The business-core route through the same chapters.

SEGMENT	LIKELY NEED	VALIDATION FOCUS	ADOPTION OPPORTUNITY
Accounting information systems	Data meaning, controls, reporting and integrity.	Which modules fit the existing course.	Modular adoption or a companion pathway, subject to course fit.
Business analytics foundations	Data preparation, trustworthy SQL measures, and reporting.	Required design depth, analytical prerequisites, and reporting platform.	A selected chapter and lab route that supplies the database foundation for analytics.

Student fees and an adoption scenario

Price is part of the adoption conversation. Current listings show the spread between a digital book, a printed text and a courseware package. The scenario beside them is simple arithmetic to frame the conversation, not a forecast.

EXISTING PRODUCT / FORMAT	LISTED U.S. PRICE	ACCESS OR FOR
Database Systems 15e eBook	US$71.99	Access through January 22, 20
Database Systems 15e eBook	US$98.99	Access through July 21, 2027
Database Systems 15e paperback	US$272.95	Print purchase
MindTap for Database Systems 15e	From US$149.00	Digital coursev product / term selection appli

ADOPTING INSTITUTIONS	SECTIONS × STUDENTS	ANNUAL SEATS
10 Recruit a small set of suitable courses; establish tools and sample-unit fit.	2 × 30	**600**
25 Confirm repeat adoption, instructor support needs and section enrollments.	2 × 30	**1,500**
50 Validate capacity, delivery costs, distribution and portfolio fit.	2 × 30	**3,000**

U.S. list prices retrieved October 25, 2026. Formats, access periods and institutional agreements affect what students pay. Adoption scenarios are arithmetic examples, not forecasts. No student price has been set for this proposal.

MAIN CONCLUSION

The evidence supports exploring two defined course routes: the applied business database / MIS course, which this book leads with, and the required business-core data / IS course. Their outcomes, tools and switching criteria should be validated separately.

02

RESEARCH 02 · SYLLABI AND INSTITUTIONS

Syllabi and institution research

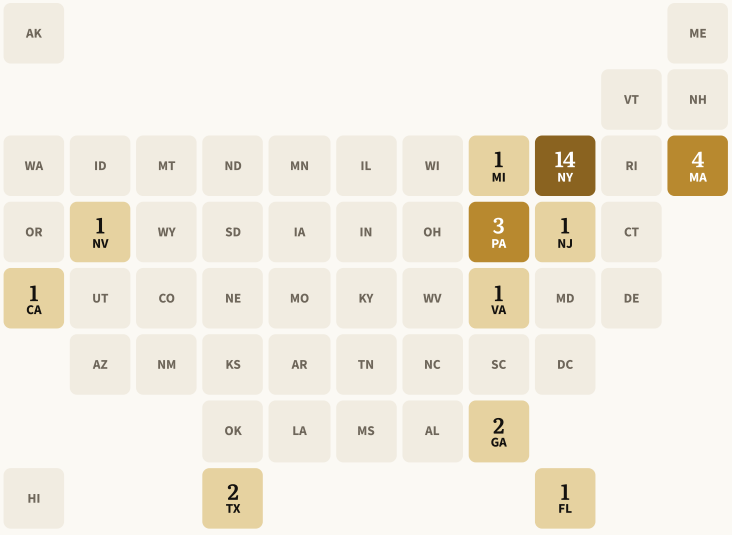
What do database and business-data courses across the country actually teach?

Full curriculum research · PDF ↗

Why and how the syllabus research was done

Why	To check the outline against what courses across the country actually teach: which topics, which tools, which textbooks, how they assess, and what they say about AI.
What we collected	63 syllabi and course documents: syllabi, catalog descriptions and course schedules for database and business-data courses, mostly undergraduate, from the public websites of 31 universities. Retrieved October 1, 2026.
63 documents, 31 institutions	Many institutions contribute more than one course, and some courses have more than one document. The 63 documents describe 54 distinct courses at 31 institutions. Of the 54 courses, 19 are business-database or business-core data courses, the closest match to this book.
How we read them	Each document was reviewed for topics, platforms, textbooks, assessment and AI policies. [S1]

The 31 institutions analyzed



14 **New York.** 8 SUNY campuses, plus Baruch, Brooklyn College, City Tech, NYU Stern, Syracuse and Hofstra.

12 **East Coast.** Rutgers, Temple, Penn State, Lafayette, Boston University, Northeastern, both UMass campuses, George Mason, Georgia, Georgia Tech and FIU.

5 **National.** UT Austin, UT Dallas, Michigan-Dearborn, San Jose State and UNLV.

D direct business-database course BC business-core data course. Unbadged courses are technical, follow-on or graduate comparisons. Together, the badged courses are the 19 closest comparators.

New York: SUNY

8 institutions

University at Albany	CSI 508
University at Buffalo D BC	MGS 404, 351
SUNY Oswego D	ISC 329
SUNY Canton D BC	CITA 215, 104
SUNY Tompkins Cortland BC	CAPS 131
SUNY Farmingdale	BCS 260, 360, 390
SUNY New Paltz	CPS 593
Stony Brook University	CSE 305

New York: CUNY and private

6 institutions

Baruch College, CUNY D	CIS 3400, 4400
NYU Stern D	TECH-UB 26
Brooklyn College, CUNY BC	CISC 1050
Syracuse University BC	IST 195
CUNY City Tech	CST 1204, 4724
Hofstra University	CSC 056

Massachusetts

4 institutions

UMass Dartmouth D	MIS 332
Boston University	CS 460, MET CS 669
Northeastern University	CS 3200
UMass Amherst	COMPSCI 345

Pennsylvania and New Jersey

4 institutions

Temple University D	MIS 2502
Rutgers University D	33:136:470
Penn State	IST 210
Lafayette College	CS 320

South

4 institutions

George Mason University (VA) D	MIS 310
University of Georgia D	MIST 4610
Georgia Tech D	MGT 4058
Florida International	COP 4722

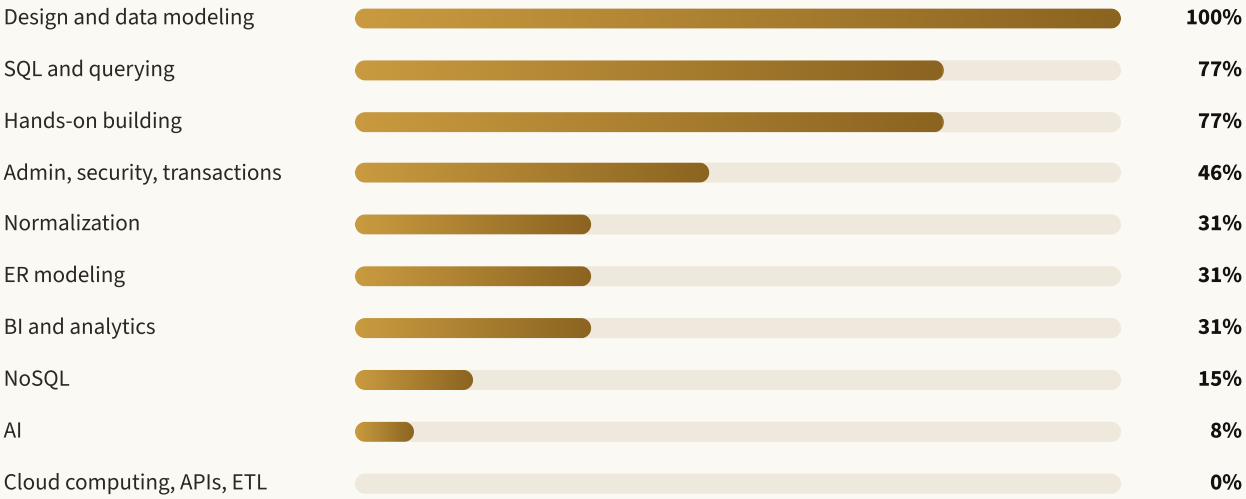
Midwest and West

5 institutions

UT Austin (TX) D	MIS 325
UT Dallas (TX) D	ITSS 4300
Univ. of Michigan-Dearborn D	MIS 321
San Jose State (CA)	INFO 242
UNLV (NV)	MIS 766

What the courses teach

Share of the course descriptions in our benchmark that name each topic. Descriptions are a lower bound: a topic a description leaves out may still be taught.



Which databases they use

Platform evidence is coded by role: **practical use** (the course assigns or demonstrates work with the tool), a **named option**, or a **software-list mention**. The chart counts any of the three, as a share of all 54 course

families and of the 19 direct and business-core families. Families can name more than one platform, and 20 of the 54 name none explicitly.



Any mention (practical use, option or listed). Among the 19 closest comparators, Access has the largest footprint (9 of 19); across all 54 families, MySQL leads (15 of 54). Practical use alone is shown in the table.

PLATFORM	PRACTICAL USE	OPTION	LISTED ONLY	COURSE FAMILIES	INSTITUTIONS	DIRECT / BUSINESS-CORE
MySQL	10	3	2	15	11	6
Microsoft Access	10	0	2	12	8	9
Oracle	4	2	2	8	7	2
Microsoft SQL Server	3	3	0	6	4	2
PostgreSQL	1	4	0	5	4	0
MongoDB	4	0	0	4	3	1
Google BigQuery	3	0	0	3	2	1
Tableau	3	0	0	3	2	1
SQLite	0	1	1	2	2	0

How they assess, and what they say about AI

Assessment Staged and team projects recur across the documents reviewed, for example at SUNY Oswego, Baruch, Boston University, Rutgers and UT Dallas, usually alongside exams.

AI	**Only 12 of the recent course offerings we reviewed include a provision for student use of AI.**

How did the findings shape the book?

Each finding, and the design decision it led to in the proposed edition.

Design and data modeling

Named in every course description in our benchmark.

→ **A two-chapter design block:** requirements to ER diagrams, then normalization.

Ch 7 · Ch 8

SQL and hands-on building

Each appears in 77% of the benchmark course descriptions.

→ **Three SQL chapters,** a build-and-test chapter, and a Let's Build in every chapter.

Ch 5 · Ch 6 · Ch 9 · Ch 10

Multiple platforms

Access has the largest documented footprint in the Direct / Business-core subset; MySQL leads across all reviewed course families. The archive also includes Oracle, SQL Server, PostgreSQL and other systems.

→ **Teach SQL in the PostgreSQL dialect,** with a portable core and explicit dialect differences; keep the existing Access route, add a planned MySQL route to be tested on the same cases, and propose Supabase as the hosted PostgreSQL environment.

App A · App B · App C

BI and analytics

Named in 31% of the benchmark course descriptions.

→ **Analytics, reporting and Power BI chapters** close the loop from query to decision.

Ch 12 · Ch 13 · Ch 14

AI and cloud

AI appears in 8% of the benchmark course descriptions; cloud, APIs and ETL in none.

→ **An AI verification from Chapter 5 on,** and a dedicated chapter on big data, data warehouses and AI.

Every chapter · Ch 15

Staged projects

Staged, multi-phase projects recur across the documents reviewed.

→ **One staged project through the semester,** with graded checkpoints and an individual demonstration at the end.

Studio A · Ch 11 · Ch 16

Selected institution evidence

A few examples show how specific courses informed the proposed learning path.

INSTITUTION	COURSE	EVIDENCE DATE / LEVEL	DOCUMENTED EMPHASIS	IMPLICATION FOR THIS PROPOSAL
Rutgers	33:136:470	Fall 2026	SQL early, design/normalization, operational topics and a staged project.	Supports use-before-design and a cumulative project.

INSTITUTION	COURSE	EVIDENCE DATE / LEVEL	DOCUMENTED EMPHASIS	IMPLICATION FOR THIS PROPOSAL
Baruch	CIS3400	Fall 2026 draft	Business requirements, ER modeling, normalization, SQL and an Access application.	Preserve application thinking and a substantive Access companion.
NYU Stern	TECH-UB26	Spring 2026	BigQuery SQL, cleaning, joins, aggregation, subqueries and window queries; ER concepts.	Supports browser/cloud SQL and an analytics-facing route.
Temple	MIS2502	Spring 2026	SQL, schemas, normalization, Python/JSON, data movement and analytics.	Connect query work to preparation and interpretation.
Georgia Tech	MGT4058	Fall 2026	Design, relational mapping, normalization, SQL and BI/warehouse concepts.	Protect the database core while connecting to reporting.
Boston University	MET-CS669	Spring 2026 · graduate	Design, SQL, operations and a project developed across six iterations.	Staged project structure is useful; graduate technical depth remains optional.
UT Dallas	ITSS4300	Spring 2026	SQL, modeling, normal forms, CTEs, windows, BI and security.	Provide advanced SQL extensions without removing essential design.
University of Georgia	MIST4610	Summer 2020	Modeling, MySQL, query work and visual answers to business questions.	Supports a query-to-communication chain; platform evidence is historical.

Inspect all 63 course documents and their findings Expand the searchable course register · online at dimapub.com

Method: 63 archived documents from 31 institutions, grouped into 54 course families. Topic shares come from the catalogue benchmark of course descriptions; the AI review covers recent primary offerings, 2024 to 2026. The searchable register keeps each source date and course role.

MAIN CONCLUSION

The courses agree on the core: design, SQL and hands-on building. They differ on platforms, and their AI policies are uneven. The book keeps the core, teaches one SQL foundation with Access and MySQL routes, and makes checking AI-assisted work a regular habit.

03

RESEARCH 03 · EMPLOYMENT AND INDUSTRY

Employment and industry research

Which of the skills this book teaches do employers ask for?

Full industry research · PDF ↗

What we researched

A 2026 analysis by 365 Data Science of 1,000 Indeed USA data analyst postings (855 relevant), U.S. Bureau of Labor Statistics projections for 2025–35, and the World Economic Forum employer survey.

What employers ask for

52.9%

OF JOB POSTINGS

U.S. data analyst job postings that ask for SQL

Of 855 relevant postings in a vendor analysis of 1,000 Indeed USA listings. Python appears in 31.2%, Power BI in 29.0% and Tableau in 26.2%.

Source: 365 Data Science [9]

+35%

GROWTH, 2025 TO 2035

Projected growth in data scientist jobs

Against 3% for all occupations. Median pay: $120,230 a year (May 2025).

Source: U.S. Bureau of Labor Statistics [10]

+9%

GROWTH, 2025 TO 2035

Projected growth in database architect jobs

Median pay: $139,500 a year (May 2025).

Source: U.S. Bureau of Labor Statistics [11]

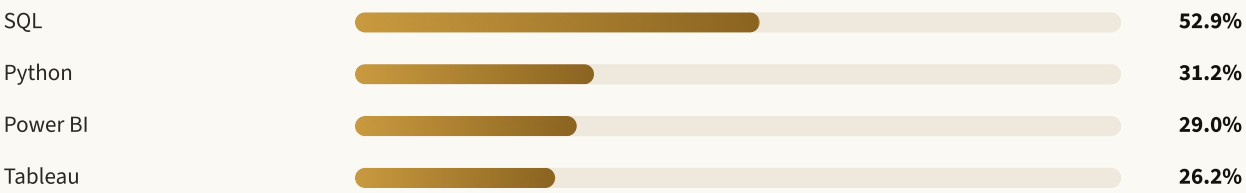
About two-thirds

OF EMPLOYERS WORLDWIDE

Plan to hire for AI skills

AI and big data are the fastest-growing skills employers report.

Source: World Economic Forum [12]



Share of 855 U.S. data analyst job postings that list each skill.

From evidence to courseware

EVIDENCE	FINDING	COURSEWARE IMPLICATION
BLS: data scientists, 2025–35	**35% projected growth**, compared with 3% for all occupations; about **24,800 openings annually**. [L1]	Connect data preparation and queries to interpretation, communication, and decisions.
BLS: database architects / administrators, 2025–35	Architects: **9%** growth. Administrators: **0%**. Combined: about **7,300 openings annually**. [L2]	Retain modeling, quality, permissions, backup, and system judgment alongside analytics.
WEF: Future of Jobs Report 2025	Survey of **1,000+ employers**, representing **14 million+ workers** across **55 economies**; roughly **7 in 10** identify analytical thinking as essential. [L3]	Require students to explain a result, test its reliability, and justify an action.
WEF: skills outlook to 2030	AI and big data lead the skills employers expect to grow fastest. [L3]	Integrate responsible AI use, verification, data provenance, and limits across the learning experience.

Job-posting shares describe this sample of 855 postings, not all U.S. hiring. BLS growth projections (the change in employment, 2025–35) and annual openings are separate measures: openings include replacement hiring, so they can be substantial even where growth is flat. Global employer expectations (WEF) are distinct from U.S. job-posting requirements.



MAIN CONCLUSION

SQL, reporting and judgment belong together. Students need to produce a result, establish that it is reliable, and communicate what an organization should do with it.

04

RESEARCH 04 · TEXTBOOKS

Textbook research

What do the leading textbooks do well, and where can this one add value?

Full textbook research · PDF ↗

How the textbook research was done

I compared the textbooks already used in these courses, starting with four especially relevant titles. The wider audit records 60 named books and course texts. I examined their chapter sequences, practice environments and course fit, then asked what this courseware should keep and where it could add value.

What the course collection tells us about textbook choice

60

TITLES

Textbooks named in the syllabi we reviewed

No title appeared at more than 3 of the 31 schools reviewed. Instructor-made materials also appear. This describes the reviewed collection, not national market share.

Source: Syllabus research, below

2021

LAST REVISION

Cengage's applied business-database title

Concepts of Database Management, 10th edition, is built on Access 2019 and SQL Server 2019. It has served the course well, and the tools and topics have moved on since.

Source: Cengage [13]

The textbooks reviewed



Database Systems: Design, Implementation, and Management

Coronel and Morris · Cengage, 15th edition (©2027)

CHAPTERS	16 in 6 parts
SQL STARTS	Chapter 4, before ER modeling
TOOLS	Auto-graded database labs in MindTap
BEST FIT	IS and MIS majors who need depth



Concepts of Database Management

Friedrichsen et al. · Cengage, 10th edition (2021)

CHAPTERS	9
SQL STARTS	Chapters 2 to 3
TOOLS	Access 2019, SQL Server 2019; MindTap and SAM
BEST FIT	Applied business courses



Database Concepts

Kroenke, Auer, Vandenberg and Yoder · Pearson, 10th edition

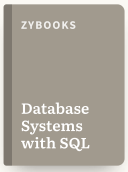
CHAPTERS	7, with more online
SQL STARTS	Chapter 3
TOOLS	Access, MySQL 8.0, Azure; running projects
BEST FIT	Short, Access-first courses



Modern Database Management

Hoffer, Topi, Venkataraman and Bala · Pearson, 14th edition

CHAPTERS	13
SQL STARTS	Chapter 5
TOOLS	Oracle, SQL Server, MySQL; an analytics chapter and State of AI sections
BEST FIT	Managerial MIS courses



Database Systems with SQL and zyLabs

Winsberg et al. · zyBooks

CHAPTERS	12 interactive chapters
SQL STARTS	Early
TOOLS	Built-in SQL sandbox; MySQL and MySQL Workbench chapters
BEST FIT	Practice-heavy courses

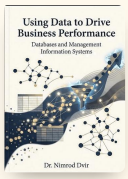


Murach's MySQL

Murach · 4th edition

CHAPTERS	Query-first workbook
SQL STARTS	Chapter 1
TOOLS	MySQL Workbench, AWS
BEST FIT	SQL skills courses

THIS BOOK



Using Data to Drive Business Performance

Dvir · proposed next edition

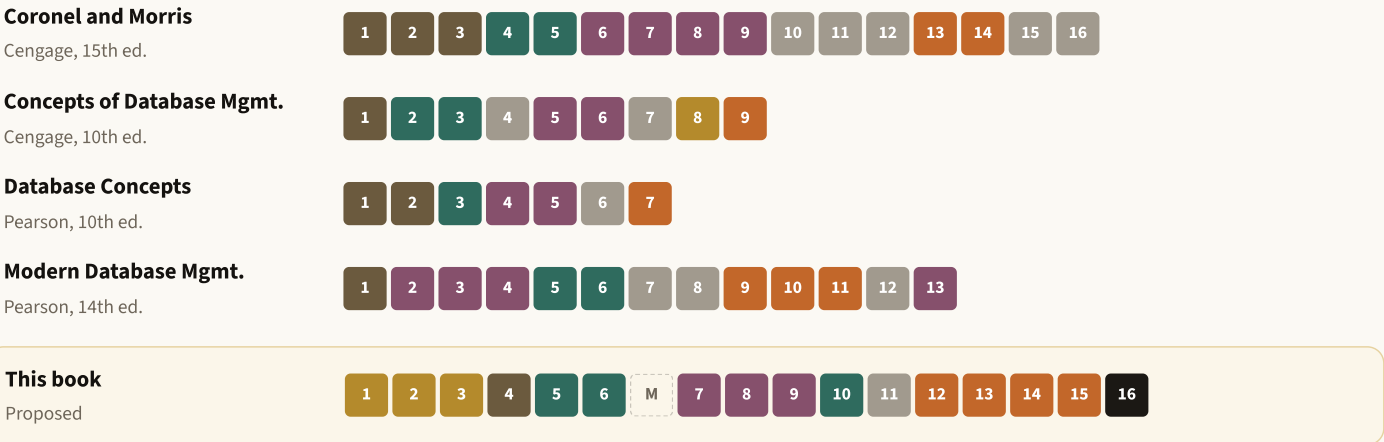
CHAPTERS	**16 in 5 parts**
SQL STARTS	**Chapter 5, after data literacy**
TOOLS	**PostgreSQL and Supabase; MySQL and Access routes; Power BI**
BEST FIT	**Business-core and MIS courses**

Publisher pages: Coronel and Morris 15e; Concepts of Database Management 10e; Database Concepts 10e; Modern Database Management 14e.
Also reviewed: zyBooks and Murach.

Chapter by chapter, in order

We coded each chapter of the four main textbooks by its main topic to see which topics the books have in common and in what order they teach them. Each square is one chapter. The last row is the proposed outline of this book.

Business and data literacy · Relational foundations · SQL · Design and building · Operations · Analytics, BI and AI · Capstone



M marks the midterm Integration Studio. Topic classification is ours, from publisher tables of contents.

What it tells us: the four compared sequences share relational foundations, SQL and design, and open with database foundations. The proposed edition keeps that core, opens with a business and data-literacy sequence, and continues into dedicated analysis, reporting, Power BI and capstone work.

What the leading books do well, and this book keeps

- **The relational core:** keys, relationships, ER modeling and normalization to third normal form
- **Early, hands-on SQL,** as in Coronel 15th edition, Kroenke and zyBooks
- **Administration, transactions and security** as a full chapter
- **Warehousing, star schemas and NoSQL** as the bridge to analytics

Where they stop, and this book continues

- **The four compared chapter sequences open with database foundations rather than a three-chapter business-performance, data-literacy and information-organization sequence.** This book does, in Chapters 1 to 3.
- **A dedicated Power BI studio is a proposed distinction in this comparison,** alongside analytics, visual reporting and a capstone decision memo. Current competing texts already cover aspects of BI and analytics.
- **Competing texts cover AI as a topic;** Modern Database Management, for example, adds State of AI sections. This design makes checking AI-assisted work a recurring practice, with structured verification activities from Chapter 5 onward.
- **Courses and books support different database products.** This proposal keeps PostgreSQL central, keeps the existing Access route, and plans a MySQL route to be tested on the same cases.

Inspect the four detailed book outlines

▼ **Database Systems: Design, Implementation, & Management** Cengage

Coronel and Morris · 15th edition; 14th retained as historical baseline [P2]

The existing course proposition

Broad database design, implementation and systems instruction. Its digital resources already include SQL practice; the current edition places SQL before formal design.

The proposed distinction

The proposed book narrows the primary audience to applied business learners and makes validation, reporting and a decision memo part of one repeated workflow.

Chapter-level sequence

1. Database systems
2. Data models
3. Relational foundations
4. SQL introduction
5. Advanced SQL

6. ER modeling
7. Advanced modeling
8. Normalization
9. Database design
10. Transactions and concurrency
11. Query performance
12. Distributed databases
13. BI and warehouses
14. Big data and NoSQL
15. Connectivity and web technologies
16. Administration and security

The 15th-edition sequence is paraphrased from the earlier contents audit; the official current listing was checked for edition and scope. The uploaded 14th edition instead puts modeling/normalization before SQL. Do not use its sequence to describe the 15th.

▼ **Concepts of Database Management** Cengage

Friedrichsen, Ruffolo, Monk, Starks, Pratt and Last · 10th edition [P3]

The existing course proposition

A concise applied introduction with QBE/algebra, SQL, normalization, relationships, operations, careers and trends. Includes cases and Access training in MindTap.

The proposed distinction

The nearest portfolio conversation: a business-performance framing with verified analytical SQL, a proposed browser-to-hosted PostgreSQL path, BI work and an integrated decision deliverable. This is a potential position, not an agreed replacement.

Chapter-level sequence

1. Introduction to database management
2. Relational model, QBE and algebra
3. Relational SQL
4. Advanced relational topics
5. Normalization
6. Relationships in database design
7. DBMS processes and services
8. Database careers
9. Industry trends

The official nine-module sequence is paraphrased. Appendices address a complete design example, SQL, FAQs, MySQL and requirements analysis.

▼ **Database Concepts** Pearson

Kroenke, Auer, Vandenberg and Yoder · 10th edition; uploaded PDF is 9th [P4]

The existing course proposition

A compact hands-on database text with multiple running examples/projects, software flexibility, cloud coverage and MySQL extensions.

The proposed distinction

The proposed distinction is the sustained business-measure-to-decision chain and explicit evidence requirements, rather than the mere existence of business cases, cloud topics or early SQL.

Chapter-level sequence

1. Getting started
2. Relational foundations
3. SQL
4. ER data models

5. Database design
6. Administration and cloud
7. BI, warehouses and big data

Seven-part chapter sequence paraphrased using publisher/reference-edition audit. The older uploaded PDF supports historical analysis and must not be labeled the current edition.

▼ **Modern Database Management** Pearson

Hoffer, Topi, Venkataraman and Bala · 14th edition [P5]

The existing course proposition

A comprehensive organizational database text. Current coverage extends from data to analysis, with an analytics chapter, notes on cloud-based solutions and new State of AI sections.

The proposed distinction

A serious comparison: the proposal must demonstrate a more bounded business-course workload and a coherent guided-to-independent learning experience; it cannot claim competitors omit analytics or AI.

Chapter-level sequence

1. Database environment and development
2. Organizational data modeling
3. Enhanced ER models
4. Logical design and relational foundations
5. SQL introduction
6. Advanced SQL
7. Databases in applications
8. Physical design and infrastructure
9. Big data
10. Integrated platforms and warehouses
11. Data-to-insight analytics
12. Administration and data quality
13. Object-oriented modeling

Current official chapter-level sequence paraphrased. Advanced normal forms and notation appear in appendices.

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Student prices for the current titles appear with the adoption scenario in the **market and adoption research**.

MAIN CONCLUSION

The leading books share a strong relational core, and this book keeps it. The difference is the frame: it opens with business performance and data literacy, ends with reporting, Power BI and a decision memo, and makes checking AI-assisted work part of the practice.

05

RESEARCH 05 · CLASSROOM

Classroom research

What happens when students actually use the book?

Open the live courseware ↗

Tested in class

The book was tested in class in summer 2026 and fall 2026, with about 600 students in total. Each round showed me where students needed more help, and the chapters were revised there.

Summer 2026

VERSION 1

Tested with a small group of students

The complete course, used end to end, then revised.

500

STUDENTS · VERSION 2

In use this fall

The revised courseware, with accounts, quiz grading and an instructor dashboard.

About 600

STUDENTS IN TOTAL

Two rounds of classroom testing

Summer and fall 2026, at the University at Albany.

> *I understand the terms, but not how they connect.*

Some version of this is what I hear from students every semester.

The syntax is *the easy part.*

The hard part is knowing which question is worth asking, getting the right data to answer it, and trusting the answer enough to put your name on a recommendation. So the book starts where business starts, with a decision someone needs to make, and students use a working database before they design one.

I write the way I teach, as a teacher talking rather than a reference manual. The chapters are revised in my own classroom, most recently after the summer 2026 test, at the places where students needed more help.

What the classroom shaped

"I understand the terms, but not how they connect." →	A course arc from data to decisions, and a business question at the start of every chapter.
Students need more context. →	Business first: Chapters 1 to 3 cover performance, data and how information is organized.
Students need more practice. →	A Let's Build, an interactive lab and an auto-graded quiz in every chapter.

Students need more confidence.	→	Use a supplied database before designing one, and learn each skill on a guided case before trying it on a second.
Trusting the answer is the hard part.	→	Checking is a habit: results checked by hand (Chapter 6), AI-written queries verified (from Chapter 5), every Power BI number validated against SQL (Chapter 14).

Classroom figures are supplied by the author. They describe development and use, not independently measured learning gains.

MAIN CONCLUSION

In my classroom, syntax is the easy part. The hard part is connecting the pieces and trusting the answer. The book is built around that connection, from the first business question to a decision students can explain and defend.

Part three

— A SUGGESTED NEXT EDITION

The proposed edition

Sixteen chapters in five parts, with a preface, a midterm integration studio, a closing chapter on careers and seven appendices, all built on the research above.

Complete 16-chapter outline · PDF ↗ | Full prospectus · PDF ↗

— AN INVITATION TO DEVELOP THIS TOGETHER

A working proposal, *open to better ideas.*

This edition is a draft and a suggestion, informed by the research and my classroom experience. I would like to work with the editors and reviewers to finalize the sequence, depth, tools and running projects. It is a starting point for our conversation.

— BUILT ON THE RESEARCH

What each study told us, and what the edition does

Not to repeat the research: the basics of each study, with a link to the full findings.

STUDY	WHAT WE LEARNED	WHAT THE EDITION DOES	
RESEARCH 01 **Market and adoption**	WHAT WE LEARNED Most business schools require an IS course, and most IS programs offer a database course.	WHAT THE EDITION DOES Two routes through one book: the full MIS database course and a shorter business-core route.	See the research ↑
RESEARCH 02 **Syllabi and institutions**	WHAT WE LEARNED Courses share design, SQL and hands-on building; platforms vary, with Access and MySQL most common; AI policies are uneven.	WHAT THE EDITION DOES A two-chapter design block, three SQL chapters, PostgreSQL as one foundation with Access and MySQL routes, and AI verification from Chapter 5.	See the research ↑
RESEARCH 03 **Employment and industry**	WHAT WE LEARNED SQL appears in 52.9% of analyst postings and Power BI in 29%; employers value analytical thinking.	WHAT THE EDITION DOES A full Power BI chapter, every number validated against SQL, and a decision memo students explain and defend.	See the research ↑

RESEARCH 04
Textbooks

WHAT WE LEARNED
The leading books share a strong relational core and open with database foundations.

WHAT THE EDITION DOES
Keeps the core; opens with business performance and data literacy (Chapters 1 to 3); ends with reporting and a capstone.

See the research ↑

RESEARCH 05
Classroom

WHAT WE LEARNED
Students understand the terms but not how they connect; trusting the answer is the hard part.

WHAT THE EDITION DOES
A course arc from data to decisions, a database used before it is designed, and checking as a habit in every chapter.

See the research ↑

WHO IT IS FOR

Course fit

Primary course	The applied business database or MIS data-management course: modeling, normalization, SQL through window functions, administration, warehousing and business intelligence, taught through business cases.
Business core	The required data or information systems course for accounting, finance, marketing, management and operations students, who need to query, check and present data. A shorter route through the same chapters serves it.
Adjacent courses	Accounting information systems, business analytics foundations, and bridge courses in IS master's programs.

HOW A CHAPTER WORKS

A familiar rhythm in every chapter

Every chapter combines the same eight components, and a semester project grows across three checkpoints. The existing **Grading Database** supports guided practice; **PetVax** lets students try the same skills independently.

01
Introduction
The business question and the chapter objectives.

02
Core concepts
The main chapter, in numbered sections with worked examples.

03
Terms Treasury
The key terms, defined in context.

04
Reading Assessment Test
An auto-graded quiz, tied to the book's own databases.

05
Let's Build
Guided, hands-on practice on the running case.

06
Interactive lab
Independent practice on a second case, with hints and verification tasks.

07
Review questions
Recall and reflection.

08
Key takeaways
A summary and a bridge to the next chapter.

Instructor materials

The courseware also includes materials for instructors:

Review questions · Terms Treasury · Answer keys · Automatic quiz grading · Rubrics · Slides · Lesson plans · Instructor dashboard

STRUCTURE, SEQUENCE AND LEARNING DESIGN

The complete chapter outline

Open a part to see every chapter: its guiding question, its sections, the guided build and lab, and the research connection behind it.

FRONT MATTER

P

Preface: Why This Book, and How It Works

Why it was written and how to use it.

- P.1 Why I Wrote This Book
- P.2 How This Book Is Different
- P.3 How a Chapter Works
- P.4 The Tools, and Why They Were Chosen
- P.5 Two Routes Through the Book
- P.6 How AI Is Used in This Book

I

PART I · CHAPTERS 1 TO 3

From Data to Decisions

Main goal: Connect data to business performance, classify data correctly and organize information.

1 From Data to Business Performance · 2 Understanding Data · 3 Organizing Data and Information

CHAPTER

1

From Data to Business Performance

How does a business turn everyday activity into better results?

- 1.1 What a Business Is and How It Creates Value
- 1.2 Business Performance, Goals and KPIs
- 1.3 From Data to Wisdom: The DIKW Pyramid
- 1.4 Information Behavior: How People Seek and Use Information
- 1.5 Information Systems and Information Technology
- 1.6 The R.E.A.D. Framework: Recognize, Extract, Analyze, Decide
- 1.7 AI in the Data-to-Decision Chain
- 1.8 The Running Cases and the Semester Project

Let's Build: Trace one KPI back to the records that produce it.

Interactive lab: A KPI audit of a small business: three KPIs, the data each needs, and who decides with it.

> **Research connection:** None of the four compared texts opens with business performance and KPIs. Business-systems framing appears in Buffalo MGS 351, Syracuse IST 195 and UNLV MIS 766.

CHAPTER
2

Understanding Data

What kind of data is this, and what can I legitimately do with it?

- 2.1 Data and Information
- 2.2 Digital and Analog Data; Bits, Bytes and Tokens
- 2.3 Data Types: Numbers, Text, Dates, Booleans and Files
- 2.4 Qualitative and Quantitative Data
- 2.5 Levels of Measurement: Nominal, Ordinal, Interval, Ratio
- 2.6 Structured, Semi-Structured and Unstructured Data: CSV and JSON
- 2.7 What Makes a Good Table
- 2.8 Data Quality
- 2.9 From Data to Insight: What Each Type Allows

Let's Build: Profile a dataset column by column: type, scale and allowed operations.

Interactive lab: Data Type Detective: classify and repair a messy business file, then draw one valid insight per column type.

> **Research connection:** The compared outlines do not give data literacy and measurement scales their own opening chapter. Data preparation appears in NYU TECH-UB 26 and Temple MIS 2502.

CHAPTER
3

Organizing Data and Information

How do we make data findable, reusable and trustworthy?

- 3.1 Why Organization Matters
- 3.2 Files, Folders, Naming and Versions
- 3.3 Hierarchies, Tags and Facets
- 3.4 Metadata, with YAML as an Example
- 3.5 Taxonomies and Controlled Vocabularies
- 3.6 Data Dictionaries
- 3.7 Spreadsheets as Tables, and Where They Break
- 3.8 From Organized Files to Databases

Let's Build: A data dictionary and a folder-and-tag scheme for the running case.

Interactive lab: Reorganize and document a shared-drive mess.

> **Research connection:** Added by design from information science, so students see why databases exist before they build one. Not a standalone topic in the syllabi reviewed.

PART II · CHAPTERS 4 TO 6

II Databases and SQL

Main goal: Read a relational database and answer business questions with SQL.

4 Databases and the Relational Model | 5 SQL Foundations | 6 SQL for Business Analysis | M Integration Studio A

CHAPTER 4

Databases and the Relational Model

Why do organizations keep data in related tables instead of files?

- 4.1 What a Database Is and What a DBMS Does
- 4.2 Local and Cloud Databases: SQLite, PostgreSQL, Supabase, MySQL, Access
- 4.3 Tables, Rows, Columns and Data Types
- 4.4 Primary, Foreign and Composite Keys
- 4.5 Relationships: One-to-One, One-to-Many, Many-to-Many
- 4.6 Schemas, Integrity Rules and Constraints
- 4.7 Database Objects: Tables, Queries, Views, Forms and Reports

Let's Build: Explore a supplied case database in Supabase Studio and trace a relationship.

Interactive lab: Read an unfamiliar schema and predict the answers to five business questions before querying.

> **Research connection:** Design and data modeling are named in every course description in our benchmark; relational foundations are documented in Baruch CIS 3400, George Mason MIS 310 and SUNY Canton CITA 215.

CHAPTER 5

SQL Foundations

How do I ask one table a precise question?

- 5.1 SELECT and FROM
- 5.2 Filtering with WHERE, LIKE, IN and BETWEEN
- 5.3 Sorting, DISTINCT and LIMIT
- 5.4 Working with NULL and Dates
- 5.5 Aggregate Functions
- 5.6 Creating Tables and Inserting Data
- 5.7 Safe UPDATE and DELETE
- 5.8 AI verification: Accept, Revise or Reject an AI-Written Query

Let's Build: Answer ten business questions on the running case.

Interactive lab: The same skills on the second case, with an AI verification.

> **Research connection:** SQL appears in 77% of the benchmark course descriptions and in 52.9% of the 855 analyst postings. Early SQL in Rutgers 33:136:470, Northeastern CS 3200 and UT Dallas ITSS 4300.

CHAPTER
6

SQL for Business Analysis

How do I combine tables into trustworthy business measures?

- 6.1 INNER and LEFT Joins
- 6.2 Joining Three or More Tables
- 6.3 GROUP BY and HAVING
- 6.4 CASE Expressions for Business Categories
- 6.5 Calculated Measures: Rates, Averages and Shares
- 6.6 Grain and the Double-Count Problem
- 6.7 Checking Results by Hand

Let's Build: Build a small KPI report from joined tables.

Interactive lab: Find and fix a double-counted measure.

> **Research connection:** Joins and aggregation are where business reports most often go wrong. Taught at Rutgers, Temple, UT Austin and UT Dallas.

MIDTERM
M

Integration Studio A (midterm)

An unnumbered review week between Parts II and III.

Cumulative review of Chapters 1 to 6, a practice exam, and **Project Checkpoint 1**: proposal and data plan.

> **Research connection:** Staged, multi-phase projects recur across the course documents reviewed, so the project has graded checkpoints through the semester.

III

PART III · CHAPTERS 7 TO 11

Designing and Running Databases

Main goal: Design, normalize, build, test and secure a database of their own.

- 7 Database Design: From Requirements to ER Diagrams
- 8 Normalization
- 9 Building and Testing a Database
- 10 Advanced SQL
- 11 Database Administration, Security and Reliability

CHAPTER
7

Database Design: From Requirements to ER Diagrams

How do business rules become a database design?

- 7.1 The Systems Development Life Cycle
- 7.2 Waterfall and Agile
- 7.3 Requirements and Business Rules
- 7.4 Entities, Attributes and Relationships
- 7.5 Cardinality, Optionality and Crow's Foot Notation
- 7.6 Associative Entities and Many-to-Many Relationships
- 7.7 Drawing ERDs with Lucidchart, diagrams.net or Mermaid

Let's Build: Draw the ERD for the running case from its business rules.

Interactive lab: Design an ERD for a new business process.

> **Research connection:** Design and data modeling are named in every course description in our benchmark. ER modeling comes before normalization, as in Coronel and Morris, Kroenke and Hoffer.

CHAPTER 8

Normalization

How do we design out errors before they happen?

- 8.1 Update, Insert and Delete Anomalies
- 8.2 Functional Dependencies
- 8.3 First Normal Form
- 8.4 Second Normal Form
- 8.5 Third Normal Form
- 8.6 Decomposing and Rejoining Tables
- 8.7 When to Denormalize

Let's Build: Normalize a flat spreadsheet to third normal form.

Interactive lab: Normalize a real business form and prove no data was lost.

> **Research connection:** Normalization is core in each of the four compared texts and is documented in Baruch, Rutgers, George Mason, Georgia Tech and SUNY Farmingdale BCS 260.

CHAPTER 9

Building and Testing a Database

How do we turn a design into a working, trustworthy database?

- 9.1 Data Definition Language: CREATE, ALTER, DROP
- 9.2 Constraints
- 9.3 Loading Data from CSV
- 9.4 Staging and Cleaning
- 9.5 Integrity Tests
- 9.6 Data Entry Workflows and Forms

Let's Build: Build and load the running-case database in Supabase.

Interactive lab: Build the lab case from its ERD and run an integrity test suite.

> **Research connection:** Hands-on building appears in 77% of the benchmark course descriptions. Implementation projects at Rutgers, Baruch and Boston University MET CS 669.

CHAPTER 10

Advanced SQL

How do analysts write queries that are reusable and easy to check?

- 10.1 Subqueries
- 10.2 Common Table Expressions
- 10.3 Views
- 10.4 Conditional Aggregation
- 10.5 Window Functions
- 10.6 Set Operations
- 10.7 Functions and Triggers
- 10.8 Reading a Query Plan with EXPLAIN

Let's Build: A reusable analysis layer of views and CTEs.

Interactive lab: A monthly performance report with rankings and period-over-period change.

> **Research connection:** Analytical SQL with CTEs and window functions appears in current syllabi at NYU Stern and UT Dallas.

CHAPTER
11

Database Administration, Security and Reliability

How do we keep business data safe, correct and available?

- 11.1 Roles, Users and Least Privilege
- 11.2 Row-Level Security
- 11.3 Transactions and ACID
- 11.4 Concurrency
- 11.5 Backup and Recovery
- 11.6 Indexes and Performance
- 11.7 Governance, Privacy and Compliance
- 11.8 Audit Trails

Let's Build: Add roles, row-level security and an audit trigger to the running case.

Interactive lab: A security and recovery plan for the lab case. Project Checkpoint 2: working database.

> **Research connection:** Administration, security or transactions appear in 46% of the benchmark course descriptions, including Rutgers, UT Austin, George Mason and SUNY Canton.

IV

PART IV · CHAPTERS 12 TO 15

From Data to Insight

Main goal: Analyze results, visualize them, report in Power BI and work with warehouses and AI.

12 Business Analytics and Decision-Making · 13 Reporting and Data Visualization · 14 Power BI · 15 Big Data, Data Warehouses and AI

CHAPTER
12

Business Analytics and Decision-Making

What does the data actually tell us, and how sure are we?

- 12.1 Four Kinds of Analytics
- 12.2 Framing an Analytical Question
- 12.3 Distributions, Segments and Fair Comparisons
- 12.4 Correlation and Causation
- 12.5 Uncertainty
- 12.6 From Measure to Decision
- 12.7 The Decision Memo

Let's Build: A SQL analysis that answers one management question.

Interactive lab: An analytical memo for the lab case.

> **Research connection:** Closes the loop from query to decision with a decision memo. Business analysis tasks appear in Temple MIS 2502 and Georgia MIST 4610.

CHAPTER
13

Reporting and Data Visualization

How do we show results so people decide well?

- 13.1 Reports and Dashboards
- 13.2 Choosing the Right Chart
- 13.3 Honest and Misleading Visuals
- 13.4 Accessible Design
- 13.5 Designing for Decision-Makers
- 13.6 The Tableau Route

Let's Build: Redesign a misleading dashboard.

Interactive lab: A one-page dashboard specification.

> **Research connection:** Only 31% of the benchmark course descriptions include BI or analytics. Visualization appears in Georgia MIST 4610 and UT Austin MIS 325.

CHAPTER
14

Power BI

How do we turn a database into an interactive business report?

- 14.1 Connecting to a Database
- 14.2 Shaping Data with Power Query
- 14.3 The Data Model and Relationships
- 14.4 Measures, Calculated Columns and DAX Basics
- 14.5 Building an Interactive Report
- 14.6 Validating Every Number Against SQL

Let's Build: A Power BI report on the running case.

Interactive lab: A Power BI report on the lab case, validated against SQL.

> **Research connection:** Power BI is listed in 29% of the 855 analyst postings and is not documented in any of the course families reviewed, so it is added by design.

CHAPTER
15

Big Data, Data Warehouses and AI

How do organizations work with data at scale, and where does AI fit?

- 15.1 OLTP and OLAP
- 15.2 Warehouses, Marts, Lakes and Lakehouses
- 15.3 Star Schemas: Facts and Dimensions
- 15.4 ETL, ELT, APIs and JSON
- 15.5 Big Data and NoSQL
- 15.6 AI and Data: Text-to-SQL, Embeddings and Retrieval
- 15.7 Verifying AI Output

Let's Build: A small star schema from the running case.

Interactive lab: Evaluate three AI answers about the lab data: accept, revise or reject.

> **Research connection:** The benchmark course descriptions rarely name AI and never name cloud computing. Warehousing appears in Baruch CIS 4400 and graduate AI topics in UT Dallas BUAN 6V99; these inform extensions, not universal requirements.

PART V · CHAPTER 16 AND CLOSING

V Putting It to Work

Main goal: Deliver a capstone decision package and plan a path into data careers.

16 Capstone: From Data to Decisions | C Where Data Can Take You

CHAPTER 16

Capstone: From Data to Decisions

Can you take a real business problem all the way to a defensible decision?

- 16.1 Scoping the Problem and the KPI
- 16.2 From Data Dictionary to ERD
- 16.3 Building the Database
- 16.4 Analysis and a Validated Dashboard
- 16.5 The Decision Memo
- 16.6 Demonstration and Reflection

Project Checkpoint 3: the final project, with a working database, analysis, dashboard and decision memo, and an individual demonstration.

> **Research connection:** Staged projects at Rutgers, Boston University and Georgia. The individual demonstration responds to the uneven AI policies documented in the syllabi.

CLOSING C

Closing: Where Data Can Take You

Where do these skills lead?

- C.1 The Data-to-Decision Journey on One Page
- C.2 Careers in Data: Analyst, BI Analyst, Database Administrator, Data Architect, Data Engineer
- C.3 Data Skills in Every Business Major
- C.4 Building a Portfolio

> **Research connection:** The employment and industry research: data scientist jobs growing 35% and database architect jobs 9% through 2035.

Appendices

These companions keep setup steps and platform alternatives close at hand without interrupting the main story. Instructors choose the tools their course needs, while students follow the same core concepts and cases.

A **Supabase and the In-Book SQL Editor**
Setting up a Supabase project, using Supabase Studio, and (planned) running PostgreSQL inside the reader.

B **MySQL Route**
Planned: schemas, seed data and solutions for every SQL chapter, to be tested on the shared cases, for courses built on MySQL.

C **Microsoft Access Route**
Tables, relationships, visual queries, forms and reports on the same cases. An Access appendix and Access labs exist today.

D **SQL Reference and Dialect Guide**
PostgreSQL, MySQL, SQL Server and SQLite side by side.

E

Diagramming with Lucidchart

ER diagrams in Lucidchart, with diagrams.net and Mermaid as free alternatives.

F

Tableau Route

The visualization chapter's activities in Tableau.

G

Aa

Terms Treasury

The glossary of every key term in the book.

How students progress

1

Understand the data

Chapters 1 to 3: what data is, how it measures performance, and how to organize it.

2

Use a database

Chapters 4 to 6: query a supplied database before designing one.

3

Build one

Chapters 7 to 11: design, normalize, build and secure their own database.

4

Decide and defend

Chapters 12 to 16: turn results into dashboards and a decision memo they can explain and defend.

ONE CONNECTED LEARNING JOURNEY

The proposed courseware

From understanding a business question to building the database, analyzing the evidence and making a decision.

The course arc

Data → Tables → Relationships → Queries → Analytics → Decisions

Read, build, try, reflect

Each chapter introduces a business question and develops the core concepts. A **Reading Assessment Test** checks the reading, a guided **Let's Build** and an independent **interactive lab** put it to work, and review questions, terms and takeaways close the chapter.

A project that grows with the student

Two running cases connect the chapter activities. A separate semester project moves through a proposal, a working database and a final dashboard with a decision memo. **See the three checkpoints**

GUIDED BUILDING AND INDEPENDENT TRANSFER

Running projects

Every Let's Build uses one guided case and every lab uses a second case, so students learn a skill once and then transfer it. The Grading Database and PetVax are the existing running cases. A small retail business is proposed as an alternative for evaluation; it is not a completed third case.

GUIDED CASE · IN USE

Grading Database

Courses, assignments and grades offer a familiar setting, although policy complexity still needs careful scoping.

What to test: Keep as the guided case with a small, stable schema.

LAB CASE · IN USE

PetVax veterinary clinic

Customers, pets, visits, services and payments give real relationships and operating KPIs.

What to test: Whether students relate to the setting.

PROPOSED ALTERNATIVE · FOR EVALUATION

Small retail business

Orders, products, customers and revenue suit the analytics, Power BI and star-schema chapters.

What to test: Whether the extra entities overload an introductory lab.

TWO ROUTES, ONE BOOK

Routes and a teaching schedule

The same chapters serve two kinds of course. Here is how I would teach each one, as a proposal to refine with editors and adopting instructors.

Two ways to use the book

Both routes reflect the syllabus research: business-core courses emphasize data use, design and reporting, while MIS courses need the full database sequence.

Business-core data course	Chapters 1 to 8, then 12 to 14.
MIS database course	Chapters 1 and 4 to 16, with Chapters 2 and 3 as assigned reading.

A proposed 14-week schedule for the MIS route

One possible pacing for the full route. The last column names what students hand in or demonstrate that week.

WEEK	CHAPTERS	EVIDENCE OF LEARNING
1	Chapters 1 and 2	A business problem, its stakeholder, a KPI and the data it needs; a first data profile
2	Chapters 3 and 4	A data dictionary; read a supplied schema and test its integrity
3	Chapter 5	Single-table SQL and a safe correction, with an AI verification
4	Chapter 6	Joins, grouped measures and a result checked by hand
5	**Integration Studio A**	**Cumulative review and practice exam; Project Checkpoint 1: proposal and data plan**
6	Chapter 7	Requirements and an ERD
7	Chapter 8	A normalized design and a check that no data was lost

WEEK	CHAPTERS	EVIDENCE OF LEARNING
8	Chapter 9	A reproducible build, load and integrity test suite
9	Chapter 10	Reusable SQL: views, CTEs and window functions
10	Chapter 11	Access, rollback and recovery evidence; Project Checkpoint 2: working database
11	Chapter 12	An analytical question answered in a short decision memo
12	Chapters 13 and 14	A dashboard specification and a Power BI report validated against SQL
13	Chapter 15	A small star schema and an AI verification report
14	Chapter 16	Capstone demonstration: rebuild, demonstrate, explain and adapt individually; Project Checkpoint 3
Variant	**Business-core route: Chapters 1 to 8 in weeks 1 to 9, with Integration Studio A after Chapter 6; Chapters 12 to 14 in weeks 10 to 13; a dashboard and decision memo as the final project in week 14.**	

14

Why 14 weeks?

A 16-week semester usually has about 14 teaching weeks once holidays, the midterm and the final are removed. The schedule fills those weeks and leaves the rest for review and the final project. Final workload, chapter selections and pacing are to be agreed with editors and adopting instructors. The closing chapter on careers is assigned reading.

ONE SQL FOUNDATION

Technology and tools

Recommendation: make **PostgreSQL** the SQL foundation, with **PGlite** for immediate browser practice and **Supabase** for the hosted project. Use **Power BI** as the main reporting studio, **Tableau** as an alternative, and **Lucidchart** for modeling with a supported alternative. Keep the existing **Microsoft Access** route; develop and test a matching **MySQL** route.

LAYER	RECOMMENDED ROLE	CONDITION FOR ADOPTION
SQL	Main skill and reasoning language.	Portable core examples with PostgreSQL-specific syntax labeled.
PGlite	Immediate embedded practice.	Planned. Implement and test persistence, reset, accessibility and known-answer checks.
Supabase	Hosted project and operational controls.	Classroom onboarding, export/restore, permission tests and a supported exit route; no guaranteed free service claim.
Power BI	Principal BI studio in Chapter 14.	Windows authoring access or a supported institutional virtual lab.
Tableau	Equivalent alternate visualization studio.	Same data, measures and rubric; avoid two required BI tools in the standard route.

LAYER	RECOMMENDED ROLE	CONDITION FOR ADOPTION
Lucidchart	Diagramming example.	Provide a supported alternative; no presumed free-tier entitlement.
Access / MySQL	Compatibility routes. Access is a route in use (labs and an appendix); MySQL is a planned route.	Test the MySQL route on the shared cases, with equivalent core examples and instructor setup; not every student must learn both.

PGlite runs locally in the browser; Supabase is hosted PostgreSQL. Export and import must be taught explicitly. Power BI Desktop requires Windows or an institutional virtual lab. Integration, access, licensing and maintenance still need implementation decisions. **[S2] [T1] [T2] [T3]**

WHY POSTGRESQL

MySQL and Access have substantial documented footprints in the reviewed courses. The book keeps meaningful routes for both while teaching one principal SQL foundation. PostgreSQL is a design recommendation, not the dominant platform in the sample.

Companions and appendices

Setup, alternative routes and reference material for each tool

COMPANION	PURPOSE
PostgreSQL / PGlite / Supabase	Setup, workspace reset/persistence, portable SQL, explicit transfer, exports, restore and access tests.
MySQL / MySQL Workbench	A planned alternate route, to be tested with matching data, query results and dialect differences. An appendix-length syntax list alone is insufficient.
Microsoft Access	In use today. Tables, relationships, visual queries, forms, reports and an equivalent small application.
Lucidchart and alternatives	Crow’s Foot modeling, exports and review; diagrams.net/ERDPlus or Mermaid alternatives avoid a mandatory subscription.
Power BI and Tableau	A complete Power BI studio plus an equivalent Tableau reporting route; Mac access planning is explicit.
SQL reference and advanced database topics	Portable core syntax, dialect flags, triggers/procedures, deeper windows and normal forms.
Capstone, reviews and instructor resources	Milestones, tests, rubrics, solution files, slides, question banks and course-route maps.

PRACTICE IN EVERY CHAPTER

Learning and assessment

Students predict and define before execution, test keys and totals during execution, and interpret the result afterward. Reading checks, guided practice, independent labs, cumulative reviews and the final project each serve a distinct purpose.

ASSESSMENT	PURPOSE	PROPOSED EVIDENCE
Reading assessment tests	Check readiness and conceptual distinctions.	Short scenario, output-prediction and misconception items; separate formative feedback from graded evidence.
Let's Build	Guided skill development.	Worked reasoning, expected results and checkpoints.
Interactive chapter labs	Independent transfer.	SQL, numeric answers, diagrams, files and short interpretations, appropriate to the chapter.
Checking AI-assisted work	Practice judging AI-assisted work, from Chapter 5 on.	Accept, revise or reject an AI-written query or answer, with a short written reason.
Project checkpoints	Stage the semester project.	Proposal and data plan at the Midterm Integration Studio, a working database in Chapter 11, and the final project in Chapter 16.
Cumulative tests / reviews	Integrate ideas across chapters.	Unfamiliar schema or changed business rule; explain and repair.
Final project	Demonstrate a complete usable solution.	Rebuildable database, validation log, report, decision memo and individual demonstration.

One problem. Three checkpoints. A defensible recommendation.

CHECKPOINT 1

Question and requirements. A scoped business problem, stakeholder, measure, source plan and acceptance criteria. Start in the opening chapters.

CHECKPOINT 2

Design and working data. An ERD, normalization rationale, implemented schema, synthetic data, import/reject log and integrity checks.

CHECKPOINT 3

Evidence and action. Tested SQL, reconciled reporting, essential access/recovery controls, a decision memo and an individual change task.

PROPOSED RUBRIC	WEIGHT
Problem framing and requirements	10%
Data model and normalization rationale	20%
Implementation and data quality	20%
SQL correctness and verification	20%
Reporting and business interpretation	15%
Reproducibility, controls and individual explanation	15%

Assessment principle

Use known-result SQL and integrity tests where they fit; use human review for modeling, explanations and business judgment. Evaluate unfamiliar variations and require a short individual explanation. Automated feedback on SQL and lab answers is a prototype in testing.

Rubric weights are a proposal for faculty review, not a change to the live course. A compact submission includes schema.sql, seed data, queries.sql, an ERD, validation evidence, a README and a short decision memo. A live link alone is not enough, because its data may change after submission.

Part four

CURRENT STUDIES AND EARLIER WORK

The full research library

Every document behind this proposal, in one place. Start with the brief; go deeper with the four studies; the earlier work shows how the proposal developed.

START HERE

The proposal

PDF

Two-page Cengage brief

The proposition, the evidence, the proposed 16-chapter edition and the next step, on two letter pages.

Open PDF ↗

PDF

Complete courseware prospectus

Every section of the proposal, with all research records and the full outline expanded.

Open PDF ↗

PDF

Detailed 16-chapter outline

Every section, build, lab and research connection, plus technology, cases and assessment.

Open PDF ↗

PDF

This page as a PDF

The whole prospectus page, with every part of the chapter outline open, ready to print or forward.

Open PDF ↗

CURRENT STUDIES

The full studies

PDF

RESEARCH 01

Full market research

Method, sourced findings, the qualified reach proxies and adoption fit.

Open PDF ↗

PDF

RESEARCH 02

Full curriculum research

The syllabus research with its visuals and all 63 course-document records.

Open PDF ↗

PDF

RESEARCH 03

Full industry research

Job-posting, labor-projection and employer evidence.

Open PDF ↗

RESEARCH 04

Full textbook research

The comparison, the four detailed outlines and all 60 catalogue records.

Open PDF ↗

EARLIER WORK

Earlier reports and presentations

Preserved as historical work. They may contain earlier platform plans or estimates; the studies above carry the current proposal.

Strategic insight dashboard	Earlier market / positioning exhibit	Open PDF ↗
Strategic analysis	Alternate file of the same two-page dashboard	Open PDF ↗
Full earlier market research	Market analysis presentation	Open PDF ↗
Publisher acquisition dashboard	Earlier commercial and courseware case	Open PDF ↗
Product prospectus	Earlier four-page prospectus	Open PDF ↗
Cengage pitch	Earlier strategic acquisition proposal	Open PDF ↗
Detailed product prospectus	Earlier three-page prospectus	Open PDF ↗
Author presentation	Thirteen-slide book overview	Open PDF ↗

Earlier interactive research pages Ten exhibits from the early research · online at dimapub.com

SOURCES AND SCOPE

References

Market, employment and curriculum references

References

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2. Fustos, J., Morris, G., & Haga, W. (2017). Do students get enough information systems education in the business core? *Business and Management Studies, 3*(2), 1–7. Study of 61 business schools. Link
3. Data USA. (2026). *Management information systems and services* (IPEDS completions, 2024). Link

4. Hwang, D., Ma, Z., & Wang, M. (2015). The information systems core: A study from the perspective of IS core curricula in the U.S. *Information Systems Education Journal, 13*(6), 27–34. Link

5. Zheng, Y., Hameed, T., Lavoie, R., & Sendall, P. (2021). An overview of current business analytics programs across US AACSB schools. *Issues in Information Systems, 22*(2), 306–317. https://doi.org/10.48009/2_iis_2021_322-333 Link

6. Yang, S. C. (2023). The master's program in information systems: A survey of core curricula in AACSB-accredited business schools in the United States. *Journal of Information Systems Education, 34*(4), 430–440. Link

7. Data USA. (2026). *Accounting* (IPEDS completions, 2024). Link

8. AACSB International. (2026, September 9). Two institutions earn AACSB initial business accreditation, September 2026 (1,098 accredited institutions in 71 countries and territories). Link

9. 365 Data Science. (2026, April 28). *The data analyst job market in 2026: Research on 1,000 job postings* (Indeed USA; 855 relevant postings). Vendor analysis, not peer reviewed. Link

10. U.S. Bureau of Labor Statistics. (2026). *Occupational Outlook Handbook: Data scientists*. Link

11. U.S. Bureau of Labor Statistics. (2026). *Occupational Outlook Handbook: Database administrators and architects*. Link

12. World Economic Forum. (2025). *The Future of Jobs Report 2025*. Link

13. Cengage. *Concepts of Database Management*, 10th edition (2021). Link

Research archive, textbook, labor and platform references

[S1] **Author's syllabus archive and manifest**

63 public course documents (syllabi, catalog descriptions and schedules) from 31 institutions, collected from institutional websites on October 1, 2026. Mixed course levels and dates; the original source of each document is linked in the course register. Archive available on request.

[S2] **Author's database-platform evidence model**

October 17, 2026 audit of the same archive. Role-coded platform evidence (practical use, option, listed) for 54 selected course families; counts recalculated October 25, 2026. Available on request.

[S4] **Author's textbook adoption and contents audit**

October 10, 2026 report: 60 named books and course texts in the archive, with school-level mentions. Historical mentions, not market shares. Available on request.

[P2] Cengage: Database Systems, 15th edition ↗

Coronel and Morris. Official current listing checked October 25, 2026: 16 chapters in six parts, with Introduction to SQL (Chapter 4) and Advanced SQL (Chapter 5) before ER modeling; auto-graded database labs in MindTap.

[P3] Cengage: Concepts of Database Management, 10th edition ↗

Friedrichsen, Ruffolo, Monk, Starks, Pratt and Last. Official publisher search results provide the nine-module outline and Access practice description.

[P4] Pearson: Database Concepts, 10th edition ↗

Kroenke, Auer, Vandenberg and Yoder. Current publisher educator page; the uploaded PDF is the older ninth edition, not the tenth.

[P5] Pearson: Modern Database Management, 14th edition ↗

Hoffer, Topi, Venkataraman and Bala. Official contents and description checked October 25, 2026: 13 chapters, SQL from Chapter 5, an analytics chapter and new State of AI sections.

[L1] U.S. Bureau of Labor Statistics: Data Scientists ↗

Current Occupational Outlook Handbook read October 25, 2026. Projects 35% employment growth, 2025-2035; all occupations 3%.

[L2] U.S. Bureau of Labor Statistics: Database Administrators and Architects ↗
Current Occupational Outlook Handbook read October 25, 2026. Architects 9%; administrators 0%; combined 4%, 2025-2035. Rounded percentages.

[L3] World Economic Forum: Future of Jobs Report 2025 ↗
Employer expectations for 2025-2030: analytical thinking essential to about seven in ten responding companies; AI and big data lead skills expected to grow. Global employer survey, not U.S. hiring requirements.

[T1] PGlite: What is PGlite? ↗
Official documentation: embedded PostgreSQL compiled to WebAssembly; runs in the browser, with optional local persistence. It is not itself a hosted cloud database.

[T2] Supabase: Database overview ↗
Official documentation: each project provides a PostgreSQL database. Hosted operations, accounts, permissions and product limits require a separate teaching plan.

[T3] Microsoft: Get Power BI Desktop ↗
Current installation documentation, checked October 25, 2026. Windows desktop authoring creates an access requirement for Mac users.

Part five

FOR OUR NEXT CONVERSATION

Open questions and next steps

TO DECIDE TOGETHER

Open questions

Three areas to decide together.

1

Publishing arrangement and portfolio positioning

The form of the publishing relationship, and where the book sits among Cengage's current titles.

2

Development scope

The tools and routes to support, the running cases, platform fit, and the sample unit reviewers see first.

3

Responsibilities, review milestones and manuscript submission

Who does what, what reviewers receive and when, and the manuscript submission timeline.

Detailed agenda The questions raised in the earlier brief and prospectus, grouped under the three areas

1. Publishing arrangement and portfolio positioning

- **Positioning:** a new applied title, a successor to *Concepts of Database Management*, or a companion to *Database Systems* (Coronel and Morris)?
- **Primary market:** the applied MIS database course first, the business-core course, or both routes from one book?
- **Complement, not compete:** where the outline should complement Cengage's current titles, and where it should not compete.

2. Development scope, tools, cases and sample unit

- **Tools and access:** which routes to support fully (Access, MySQL, PostgreSQL), and how students reach Power BI.
- **Platform fit:** how the chapter package (reading assessments, labs, SQL practice) maps to MindTap, and whether a live SQL environment such as Supabase or in-browser PostgreSQL can connect to it.
- **Running cases:** keep the Grading Database and PetVax, or evaluate a commercial case such as retail orders and revenue?
- **Sample unit:** which complete unit (reading, guided build, lab, solution and rubric) reviewers should see first.

3. Responsibilities, review milestones and manuscript submission

- **Review package:** what reviewers receive (this outline, sample chapters, a lab, reader access), how many reviewers, and with what criteria.
- **Responsibilities:** who maintains the platform, the cases and the instructor materials during development.

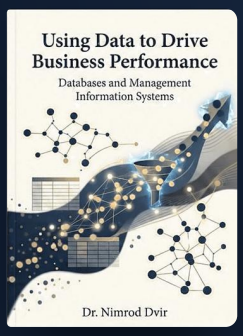
- **Timeline:** the key review dates and the manuscript submission plan.

MOVING FORWARD

Suggested next steps

I would like to formalize our publishing relationship and agree on the development scope and manuscript submission timeline.

1. **Let's cement the publishing relationship.** Agree on the arrangement and the book's place in the portfolio.
2. **Agree on the development scope.** Settle the audience, tools, cases and the sample unit.
3. **Set the milestones.** Agree on responsibilities, review milestones and the manuscript submission timeline.



CONTACT

Nimrod Dvir, PhD

Massry School of Business, University at Albany, SUNY

ndvir@albany.edu Copy email

646-244-2338 LinkedIn ↗ University page ↗

Open the live reader ↗ Read the two-page brief ↗

A FINAL THOUGHT

This book is a labor of love, and of many, many hours of work. I would be honored to see it come to life as a published book, and to see other universities and instructors make it their own.

Nim